Explanation Is the Bottleneck: Measuring and Gating XAI Cost in Co-located, Fog and Cloud Deployments of Real-Time Stress Detection

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***Abstract*—Explainable AI (XAI) is routinely attached to wearable stress classifiers, yet the time it takes to produce an explanation is almost never measured, and never inside a deployed real-time pipeline. We deploy a frozen 49,890-parameter 1D-CNN trained on WESAD as a TCP client–server system and place its server co-located with the sensor client, on a fog node and on a cloud VM. Under an open-loop sensor client with drift-corrected clock synchronisation, we measure latency, queuing delay, execution time, response time, jitter, bandwidth and resource usage. On the fog node a single region-based LIME or SHAP (GradientExplainer) explanation costs 390–1,426 ms, i.e. 111–404 × the entire server-side computation of a window without explanation. Without explanation the cloud placement is 8.69 × slower than co-location; with the most expensive explainer the gap shrinks to 1.13 ×, so explanation, not placement, determines response time. We therefore gate explanations on model confidence. At $\tau = 0.80$, measured over 1,030 windows, mean response time falls from 450.5 to 80.1 ms (5.6 ×) and single-worker capacity grows 7.5 ×, while 89.7% of misclassified windows of a well-calibrated subject are still explained. A mixture model predicts the measured mean within 2.0%. The benefit is conditional on calibration: on a second held-out subject with 0.227 expected calibration error only 12.0% of errors are caught.**

***Index Terms*—explainable AI, fog computing, real-time systems, stress detection, LIME, SHAP, selective explanation, latency measurement**

Draft built from v1 data. Tier comparisons below come from single closed-loop runs on three different software stacks (Py 3.11.9, torch 2.8.0+cpu; Py 3.14.7, torch 2.14.0+cpu; Py 3.11.16, torch 2.14.0+cu130) and a desktop fog node. They are placeholders until the v2 runs (one container image, Raspberry Pi fog node, open-loop client, 3 alternating rounds) regenerate `numbers.tex`. Remove this box for submission.

I. Introduction

Wearable stress detection on public datasets such as WESAD [1] is mature, and post-hoc explanations (LIME [2], SHAP [3]) are now a standard companion to the classifier. Explanations are, however, evaluated almost exclusively for quality: the Co-12 review of XAI evaluation explicitly excludes run time as a property that does not affect explanation quality [5], and work that applies XAI to WESAD reports neither inference nor explanation time [6]. In a real-time monitoring system this omission is not cosmetic. A new 10 s window arrives every 5 s; whatever the server spends explaining one window is time it cannot spend on the next window or on the next user.

Systems work on fog-based physiological monitoring measures exactly these transport-level quantities—latency, queuing delay, execution time, response time, jitter, bandwidth and resource usage—but without explanations [7], [8]. The two literatures have not met: XAI cost is reported without a network, and network cost without XAI.

This paper puts both in one budget. Our contributions are:

1) An end-to-end measurement of explanation cost inside a real TCP client–server stress detection pipeline at three placements, using the seven metrics of [7], an open-loop sensor client and drift-corrected clock synchronisation (Sec. III).
2) Evidence that explanation dominates the pipeline by two orders of magnitude and collapses the latency differences between placements (Sec. V-A).
3) A measured characterisation of confidence-gated selective explanation: cost, capacity and error coverage over 1,030 windows, validated against a mixture-model projection (Sec. V-B).
4) The calibration condition under which gating pays off, and the quality trade-off between four explainer operating points (Sec. V-C).

II. Related Work

Fog-based physiological monitoring. Erzurumluoğlu et al. [7] place emotion recognition on DREAMER on a Raspberry Pi 4 fog worker and report the seven metrics we adopt; their worker answers in 43.3 ms and their cloud in 73.3 ms. HealthFog [8] compares edge, fog and cloud placements for cardiac diagnosis in terms of latency and energy. Neither includes explanations. El Rashidy et al. [9] combine a fog architecture with LIME and Grad-CAM for emotion recognition but report no per-inference latency or explanation cost.

The cost of explanation. ShaTS [10] states the only explicit real-time bound we found—an explainer taking more than 1 s per window is unsuitable for real-time use—and shows KernelSHAP violating it. Reported costs vary by orders of magnitude with hardware, input size and estimator: KernelSHAP takes seconds to minutes per sample [11], [12], whereas amortised estimators such as FastSHAP [12] trade a long training phase for millisecond explanations. On constrained

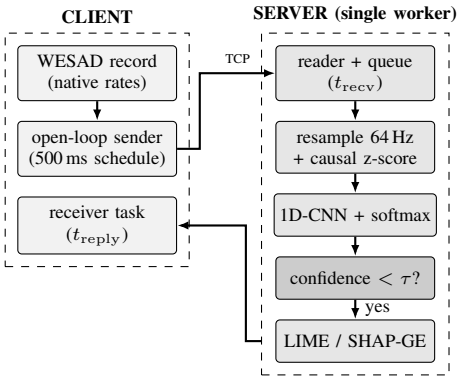


Fig. 1. Pipeline. The explanation is produced by the same worker as the prediction and inside its execution time, so its cost is visible in every metric. The server is placed co-located, on a fog node or in the cloud.

hardware, a conditionally triggered SHAP on a Raspberry Pi 4 added about 195 ms per prediction [13].

Selective computation. Gating computation on softmax confidence is established for inference (early exits [14], confidence cascades [15]) and has been proposed for explanations: budgeted selective explanations [16] and explanations of rejected samples [17]. We do not claim the mechanism; we characterise it end to end in continuous physiological monitoring, where its benefit turns out to hinge on calibration [18].

III. System and Measurement Method

A. Model

The classifier is a temporal 1D-CNN (two convolution blocks with [32, 128] filters, kernel 9, BatchNorm–ReLU–MaxPool–Dropout, global average pooling, two dense layers; 49,890 parameters) trained with focal loss [19] on 14 chest and wrist channels resampled to 64 Hz, in 10 s windows with 5 s hop. The architecture attains 0.92 F1 under 15-fold leave-one-subject-out validation; the deployed checkpoint was trained on 13 subjects with S11 and S14 held out (validation F1 0.939). We chose it over a transformer from our offline study that reached higher single-split accuracy because it is 37× smaller and 29× faster per window on the same CPU (1.86 M parameters, 103.7 vs. 3.6 ms; measured offline[TODO: archive this measurement under results/]); the model is frozen and used unchanged in every measurement.

B. Real-time pipeline

The client replays a WESAD record at the sensors' native rates (chest 700 Hz, wrist 4–64 Hz) in 500 ms ticks; each tick is one binary frame of 11,553 bytes. Resampling happens on the server and is part of the measured execution time. Offline training normalised each subject with statistics of the full record, which leaks the future; the server instead uses causal (Welford) normalisation from data seen so far. Under this constraint the deployed model reaches 0.822 accuracy and 0.694 stress F1 on 1,198 labelled windows of S11 and S14—the drop is the price of leak-free normalisation, not of the transport.

C. Transport and measurement

Single worker with an explicit queue. A reader task timestamps each frame on arrival (t_{recv}) and enqueues it; one worker drains the queue (t_{start}, t_{end}). Latency $= t_{\text{recv}} - t_{\text{send}}$, queuing delay $= t_{\text{start}} - t_{\text{recv}}$, execution $= t_{\text{end}} - t_{\text{start}}$, response $= t_{\text{reply}} - t_{\text{send}}$; jitter is the mean absolute change of consecutive response times, as in [7]. Nagle's algorithm is disabled (`TCP_NODELAY`).

Open-loop load. A client that waits for each reply before sending the next tick is a *closed* load generator: when an explanation takes longer than the 500 ms tick, the emulated sensor stops, later ticks are sent in a burst, and the waiting is omitted from the measurement—the coordinated-omission problem of closed-loop benchmarking [20]. In a local check with the most expensive explainer, a closed-loop client stalled the sensor stream by up to 661 ms (95th percentile); our open-loop client sends every tick on its schedule from a separate task (send-lag p95 20 ms, bounded by the Windows timer). We additionally report the *sensor age* of a decision, t_{reply} minus the scheduled send time, which is immune to coordinated omission.

Clocks. One-way latency needs the client and server clocks. We use monotonic `perf_counter` clocks and estimate their offset NTP-style from the minimum-RTT exchange of 12 ping rounds [21]. Because crystal drift made a single estimate yield negative latencies over 87-minute runs, the offset is re-measured every 60 s and interpolated linearly per packet. The uncertainty is half the minimum RTT; where it is of the same order as the latency itself (the cloud), we report the network share $(\text{response} - \text{execution} - \text{queuing})/2$ instead of one-way latency.

D. Explainers and confidence gating

Region-based LIME. The 14×640 window is divided into 20 time segments per channel (280 regions); a masked region is replaced by the channel mean, samples are weighted with an exponential kernel and fitted with ridge regression. Operating points: 500 samples (*LIME*) and 250 (*LIME-cheap*). **SHAP-GE.** SHAP's GradientExplainer (expected gradients [4]) with a fixed 100-window k-means background drawn from training subjects in the causal input space; 200 samples (*SHAP*) and 50 (*SHAP-cheap*). This is not KernelSHAP, whose cost structure differs [11]. Explainers are warmed up at server start-up (the first SHAP-GE call takes seconds), outside measurement.

Gating. Let c be the maximum softmax probability. A window is explained iff $c < \tau$; $\tau = 0$ never explains and $\tau = 1$ always does.

IV. Experimental Setup

Placements (Table I). We use *co-located* rather than "edge": client and server run as separate processes on the same machine and talk over loopback, which is the lowest-latency placement available; it is not a wearable. The fog server is a separate device on the local network and the cloud server a VM reached over the Internet. All three servers run the same container image (Python 3.11, PyTorch CPU build), so

TABLE I
PLACEMENTS. NETWORK SHARE IS (RESP. − EXEC. − QUEUE)/2 WITHOUT EXPLANATION (MEDIAN).

Tier	Server hardware	Software stack	Net
Co-loc.	Intel x86-64, 20 cores, 16 GB; loopback	Py 3.11.9, torch 2.8.0+cpu	0.7 ms
Fog	Intel x86-64, 12 cores, 16 GB; LAN	Py 3.14.7, torch 2.14.0+cpu	3.5 ms
Cloud	DO-Regular, 1 core, 2 GB; Internet	Py 3.11.16, torch 2.14.0+cu130	29.7 ms

the tiers differ only in hardware and network; the fog node is a Raspberry Pi 4 on Ethernet with active cooling, the performance governor and throttling flags logged per session.

Protocol. Subject S11 (held out), causal normalisation, one PyTorch thread, 15 warm-up windows excluded. Each (placement, explainer) cell is run three times in alternating order so that thermal and frequency drift spread evenly across placements; the unit of replication is the run, and we report the median over runs of per-run window medians and 95th percentiles. Resource figures are per session. Single-worker capacity is the 5 s hop divided by mean execution time—an upper bound that ignores network and phase alignment of clients.

V. RESULTS

A. Explanation dominates the pipeline

Without explanation, the fog server spends a median 3.5 ms per window on resampling, normalisation and inference—0.07% of the 5 s hop. A single explanation costs 390 to 1,426 ms (Table II), 111–404 × the whole pipeline it explains, and response time grows 137 ×. Explanation accounts for nearly all of the execution time once enabled, and peak server memory more than doubles because the explainer's background set and graph live alongside the model. Against the 1 s/window bound of [10], both cheap operating points fit on the desktop-class servers while the full LIME and SHAP-GE points exceed it at the 95th percentile.

Placement under explanation. Without explanation, placement dominates response time: the cloud is 8.69 × slower than co-location because its network share (29.7 ms) is an order of magnitude larger than the computation. With explanation the ordering is set by compute: cloud/co-located is 0.93 × for SHAP-cheap and 1.13 × for SHAP, because tens of milliseconds of network vanish next to a second of explanation. [TODO: v2: rewrite this paragraph from the Pi numbers. Expected: the Pi fog is compute-bound, every-window explanation exceeds the 1 s bound and may approach the 5 s hop, so placement matters again—through compute, not network.] In v1 the fog/co-located ratio of explanation time *changes sign* across explainers (1.69 for LIME, 0.88 for SHAP-cheap), which hardware alone cannot cause; it reflects the different PyTorch builds and is why v2 uses one image.

We also observe that the network share of response time grows when explanations are enabled on wireless and Internet paths (fog 3.5 → 24.2 ms, cloud 29.7 → 43.1 ms with SHAP-cheap) but not on loopback. A plausible cause is radio power saving during the idle gap while the server explains; the Ethernet-connected fog node in v2 separates the two.

B. Selective explanation recovers the cost

We ran the full 87-minute S11 record on the fog node with SHAP-cheap and $\tau = 0.80$ (Table III). 145 of 1,030 windows (14.1%) were explained at 365 ms each; mean response fell from 450.5 to 80.1 ms (5.6 ×) and capacity rose from 12 to 90 clients per worker (7.5 ×). The median window is not explained (11.0 ms) and the 95th percentile is an explained window (446 ms). The bursty load did not build a queue: the 99th percentile queuing delay of prediction packets stayed at 0.06 ms. A mixture model that combines the calibration-set explanation rate with the per-explanation cost predicted the mean response within 2.0% and the explanation rate within 2.3%, so the projections for other explainers and thresholds can be trusted.

Short runs are misleading here: in the first five minutes of S11 the lowest confidence was above 0.9, and no threshold triggered a single explanation. Confidence is non-stationary within a record, so gating must be evaluated on full records.

C. The calibration condition

Gating saves cost on both held-out subjects, but it only targets errors if confidence is calibrated [18]. The model is equally confident on both subjects (mean 0.909 vs. 0.908) but correct on 0.952 vs. 0.690 of windows; S14 is confidently wrong (Table IV). At $\tau = 0.80$ the explanation rate is similar (14.4 vs. 11.8%), yet 89.7% of S11's errors and only 12.0% of S14's are explained. Cross-subject temperature scaling made both worse (ECE 0.256 and 0.276) because the miscalibration has opposite signs. The cost claim and the coverage claim must therefore be stated separately: the former holds generally, the latter requires per-user calibration.

D. Explainer quality versus cost

No operating point dominates on faithfulness: all differences lie within one standard error (Table V). The explainers separate on stability instead: LIME is 2.7 × less sensitive to small input noise than SHAP-GE, while SHAP-cheap is the cheapest point. The choice is therefore between cost and stability, not fidelity, and the cheap points lose nothing measurable in faithfulness.

VI. DISCUSSION AND LIMITATIONS

Where the architectural decision moves. In the reference fog study [7], as in our runs without explanation, the placement decision is a latency decision. Once explanations are produced, the decisive parameter becomes the triggering policy: it changes mean response time by 5.6 × and capacity by 7.5 ×, more than any placement change we measured.

Comparison with the reference system. Only bandwidth is directly comparable, because hardware and windowing differ: streaming raw multichannel signal costs us 187 kbps, versus

TABLE II

RESPONSE TIME AND EXPLANATION COST PER PLACEMENT AND EXPLAINER (MS; MEDIAN AND 95TH PERCENTILE OVER WINDOWS; XAI = MEDIAN EXPLANATION TIME INSIDE EXECUTION). SINGLE RUNS OF 44 WINDOWS (V1).

	Co-located			Fog			Cloud		
Explainer	resp. p50	p95	xai	resp. p50	p95	xai	resp. p50	p95	xai
none	7.3	9.7	–	10.6	42.2	–	63.5	154.5	–
SHAP-cheap	462.0	511.1	445	442.6	574.0	390	429.8	509.0	340
LIME-cheap	352.8	585.0	346	645.2	802.5	584	463.3	608.3	340
LIME	690.0	1,177.0	682	1,227.5	1,407.1	1,154	870.1	1,091.8	744
SHAP	1,154.5	1,833.3	1,143	1,456.2	1,548.5	1,426	1,300.5	1,715.8	1,169

Server memory (peak RSS, MB): none / SHAP — co-located 187 /523; fog 210 /464; cloud 518 /838.

TABLE III

NEVER / SELECTIVE / ALWAYS EXPLAINING ON THE FOG NODE (SHAP-CHEAP). MEANS ARE REPORTED BECAUSE THE SELECTIVE DISTRIBUTION IS BIMODAL.

Policy	**Win.**	**Expl.**	**Resp.** mean (ms)	**Exec.** mean (ms)	**Clients** per worker
never (τ=0)	44	0%	15.9	3.59	1,394
selective (τ=0.80)	1,030	14.1%	80.1	55.3	90
always (τ=1)	44	100%	450.5	400.2	12

TABLE IV

EXPLANATION RATE AND SHARE OF MISCLASSIFIED WINDOWS THAT ARE EXPLAINED, PER THRESHOLD (CAUSAL INPUT SPACE, UNCALIBRATED SOFTMAX).

	S11 (ECE 0.047)		**S14** (ECE 0.227)	
τ	explained	errors caught	explained	errors caught
0.70	10.4%	62.1%	4.4%	6.0%
0.80	14.4%	89.7%	11.8%	12.0%
0.90	21.4%	100.0%	29.3%	28.8%

373.9 kbps for sending 24 extracted features at 32 requests per second in [7].

Limitations. (i) Two held-out subjects; the calibration finding is an observation on $n = 2$, not a generalisation. (ii) One worker per server; multi-worker scheduling is future work. (iii) Capacity is an analytical upper bound; multi-client open-loop runs are in progress. (iv) No user study of explanation usefulness. (v) This is an emulation: recorded data replayed at native rates, not a live wearable.

VII. CONCLUSION

Measured end to end, explanation—not inference and not placement—is the bottleneck of real-time explainable stress detection: it costs two orders of magnitude more than the pipeline it explains. Confidence-gated explanation recovers most of that cost and raises single-worker capacity 7.5 ×, but explains errors only where the model is calibrated. We release the measurement harness, including the open-loop client and drift-corrected clock synchronisation. [TODO: repository link]

TABLE V

EXPLAINER OPERATING POINTS: FOG COST AND QUALITY ON 40 WINDOWS OF S11 (FAITHFULNESS = INSERTION−DELETION AUC ± S.E.M.; LOWER SENSITIVITY IS BETTER).

Explainer	**xai (ms)**	**Faithfulness**	**Monoton.**	**Sensitivity**
SHAP-cheap	390	0.250 ±0.043	0.565	0.123
LIME-cheap	584	0.209 ±0.035	0.604	0.049
LIME	1,154	0.232 ±0.037	0.579	0.045
SHAP	1,426	0.263 ±0.044	0.549	0.128

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